National University of Computer and Emerging Sciences, Islamabad Campus

Course:	Data Structure	Course Code:	CS-2001
Program:	BS(Computer Science)	Semester:	Fall 24
Duration:	10 Minutes	Total Marks:	15
Paper Date:	28-08-2025	Name:	M. Rehaan
Section:	A		
Exam:	Quiz - 1	Roll No:	22I-2327

Problem

You are given multiple code snippets. Assuming there are no syntax errors in any of the snippets, please answer the question provided under each snippet. Justify your answer with a brief explanation, no longer than two lines.

Snippet -1:

```
#include <iostream>
using namespace std;

template <class T, int max>
int arrMin(T arr[], int n)
{
  int m = max; // 10,000  //256
  for (int i = 0; i < n; i++)
    if (arr[i] < m)
      m = arr[i];

  return m;
}

int main()
{
  int arr1[] = {10, 20, 15, 12};
  int n1 = sizeof(arr1)/sizeof(arr1[0]);  // 4

  char arr2[] = {1, 2, 3};
  int n2 = sizeof(arr2)/sizeof(arr2[0]); // 3

  cout << arrMin<int, 10000>(arr1, n1) <<endl; //10
  cout << arrMin<char, 256>(arr2, n2); // 1
  return 0;
}
```

4+1

Output: 10 1

Justify: ~~each time~~ first time when loop start it updates the value of m then compares other values with the first updated value o

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Snippet -2:

```
#include <iostream>
using namespace std;

template <typename T>
T max(T x, T y)
{
  return (x > y)? x : y;
}
int main()
{
  cout << max(3, 7) << std::endl; // 7
  cout << max(3.0, 7.0) << std::endl; // 7.0
  cout << max(3, 7.0) << std::endl; //
  return 0;
}
```

X

Output: 7 7.0 7.0

Justify: As the function max returns the greatest value.

Snippet -3:

```
#include<iostream>
#include<stdlib.h>
using namespace std;

template<class T, class U>
class A {
  T x;
  U y;
  static int count;
};

int main() {
  A<char, char> a;
  A<int, int> b;
  cout << sizeof(a) << endl; // 2
  cout << sizeof(b) << endl; // 8
  return 0;
}
```

5

→ must mention role of static var.

Output: 2 8

Justify: Because char has size of 1 byte and int has size of 4 bytes.

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